

# Optics System Design of Microwave Imaging Reflectometry for the EAST Tokamak\*

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**Abstract** A front-end optics system has been developed for the EAST microwave imaging reflectometry for 2D density fluctuation measurement. Via the transmitter optics system, a combination of eight transmitter beams with independent frequencies is employed to illuminate wide poloidal regions on eight distinct cutoff layers. The receiver optics collect the reflected wavefront and project them onto the vertical detector array with 12 antennas. Utilizing optimized Field Curvature Adjustment lenses in the receiver optics, the front-end optics system provides a flexible and perfect matching between the image plane and a specified cutoff layer in the plasma, which ensures the correct data interpretation of density fluctuation measurement.

**Keywords:** density measurement, multi-dimension, microwave diagnostic, optics lens

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(Some figures may appear in colour only in the online journal)

## 1 Introduction

High spatial and temporal resolution imaging diagnostic techniques are important for plasma transport and micro-turbulence studies in magnetically confined plasma. 2D imaging of electron temperature fluctuations has been achieved by Electron Cyclotron Emission Imaging (ECEI) on several tokamak devices [1–4], which is immensely helpful in the investigation of important issues like MHD and heat transport. On the other hand, the imaging of density fluctuations is another important measurement for particle transportation [5–8]. The typical density fluctuation structure not only has radial wavenumbers, but also has significant poloidal wavenumbers, which means the one-dimensional geometric assumption breaks down. In order to keep the simple data interpretation method valid, where the density fluctuation is proportional to phase variation, imaging optics with large aperture are employed to collect the reflected wavefront onto the detector array to reduce the interference pattern caused by the curved unsmooth cutoff layer [9,10]. The ideal transmitter and receiver optics should generate a curved wavefront or image plane to match the cutoff layer perfectly. Therefore, the design of the front-end optics is pivotal for Microwave Imaging Reflectometry (MIR).

In this paper, the details of the front-end optics system for the MIR diagnostic on EAST tokamak will be presented. The optimized optics system has the capability to adjust the shape of the wavefront over a wide

range to match the target by moving field curvature adjustment (FCA) lenses.

## 2 EAST MIR optics system

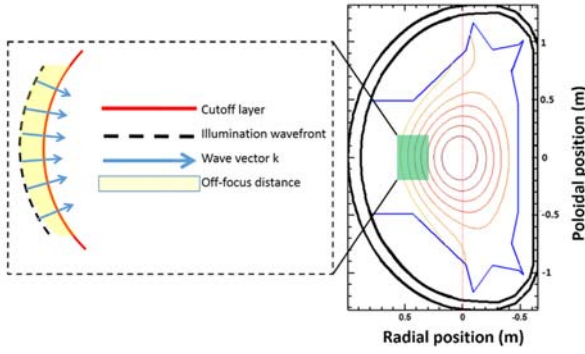
The Microwave Imaging Reflectometry system for EAST, which is the first 3rd generation MIR system employing FCA optics and fast tunable frequency probe beams, is a major advance over previous systems to provide time resolved 2D density fluctuation images. In the design, the simulations are based on the H-mode parameters on DIII-D with 1 MA plasma current and typical 2.0 T toroidal magnetic field.

### 2.1 Transmitter optics in MIR

As an active diagnostic, the MIR system needs to provide incident beams to illuminate the cut-off layers. The transmitter horn launches an ingoing wavefront to illuminate cutoff layers located over a wide radial range from normalized radius 0.2 to 0.95. The illumination wave propagates along the optical path through the transmitter optics system. The radius of the wavefront is adjustable to match the cutoff layer on the plasma side in different magnetic configurations by moving the E lens, which stands near the launch horn. It helps to keep the wave vector perpendicular to the cutoff layer over the whole illuminated poloidal region as shown in Fig. 1. The beam radius on the cutoff layer is related to the wavefront radius. Both of the transmitter and

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receiver optics share the zoom part, which is the front part of the whole optics system.



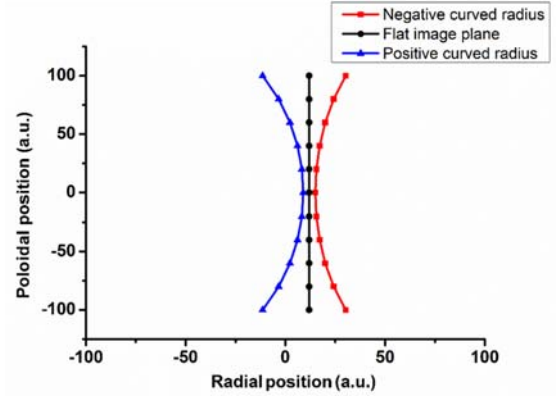
**Fig.1** Transmitter wavefront is designed to match the cut-off layer in plasma

## 2.2 Receiver optics in MIR

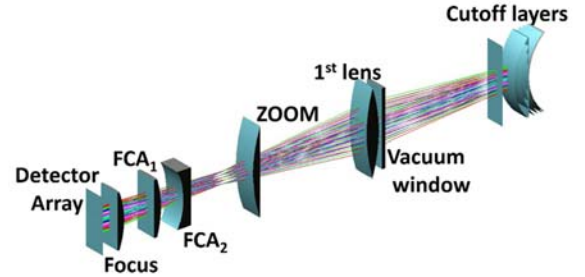
The receiver optics is able to focus on the illuminated area by the transmitter wave to obtain the scattered wavefront reflected from the cutoff layer. The collected emission then projects on an antenna array with 12 detectors<sup>[11]</sup> aligned in the vertical direction. Without any field curvature adjustment lenses, the image plane of the large aperture optics system is a curved line as indicated by the blue line shown in Fig. 2. The curved radius is a positive number. Unfortunately, the curved radius of the cutoff layer is in the reversed direction, which has a negative radius. Consequently, the usage of field curvature adjustment lenses plays the key role in MIR receiver optics to adjust the shape of the image plane into a curved surface with the same negative radius as the cutoff layer on the plasma side. In fact, the FCA lenses are also adopted in the ECEI diagnostic to correct the shape of the ECEI image plane, which is helpful to reduce overlapping between neighboring channels and obtain more precise fluctuation contours in the core plasma region.

As shown in Fig. 3, the receiver optics system of the EAST MIR adopts 5 lenses in the optics path including the 1st lens, zoom lens, and FCA lenses and the focus lens. The reflected wavefront will be collected by them through the plasma and vacuum window and then focus on the detector array. The key parameters of the receiver lenses are shown in Table 1. All lenses are made of HDPE, which has a constant refraction index in the frequency range of the MIR operation region according to the testing result. The optics can be divided

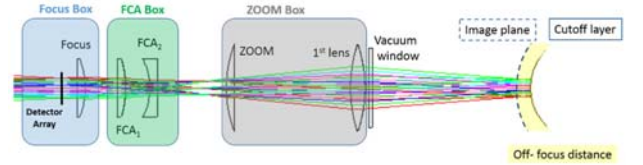
into three parts as shown in Fig. 4 with independent functions as follows: Zoom box, FCA box, and Focus box. The performance of receiver optics depends on the setup of these three specific boxes.



**Fig.2** Different shapes of image planes of large aperture image optics with negative radius (red), positive radius (blue) or infinity radius (black)



**Fig.3** 3D model of MIR receiver optics. The reflected wavefront from different poloidal positions will be imaged on a different detector on the detector array



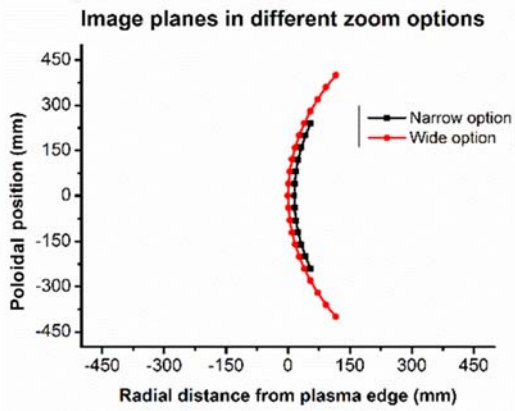
**Fig.4** The receive optics is divided into three boxes: gray zoom box, green field curvature adjustment box and light blue focus box

In the gray Zoom box, the 1st lens stands close to the vacuum window for collecting as much of the scattering wavefront as possible into the receiver optics path. Considering the space limitations, the largest aperture in the poloidal direction is 990 mm for the maximum transmission rate. The hyperbolic shape has been used

**Table 1.** The key parameters of lenses in EAST MIR receiver optics

| Lens name  | Surface type | Radius (Left) | Radius (Right) | Thickness | Refraction index |
|------------|--------------|---------------|----------------|-----------|------------------|
| 1st lens   | hyperbolic   | 1650 mm       | -1650 mm       | 180 mm    | 1.58             |
| Zoom lens  | Y Toroid     | 1500 mm       | Infinity       | 100 mm    | 1.58             |
| FCA2 lens  | Y Toroid     | -900 mm       | 4500 mm        | 100 mm    | 1.58             |
| FCA1 lens  | parabolic    | Infinity      | -1200 mm       | 100 mm    | 1.58             |
| Focus lens | Y Toroid     | Infinity      | -900 mm        | 100 mm    | 1.58             |

on both sides of the 1st lens with a -3 conic constant. It helps to reduce field curvature with positive radius, but not enough to generate the image plane with negative radius. The size of the observation area in the poloidal direction can be changed by the zoom lens to achieve different spatial resolution. There are two zoom options. For the narrow option, the zoom factor is about 0.85, which has about a 244 mm poloidal observation range with an 18.8 mm resolution. In the wide zoom option, the zoom factor increases to 1.4 by increasing the distance between the fixed 1st lens and the moveable zoom lens. The largest measurable range in the poloidal direction is 400 mm with 30.8 mm resolution. The poloidal observation range and spatial resolution depend on the zoom box performance. Fig. 5 shows both narrow and wide zoom options on the same cutoff layer.

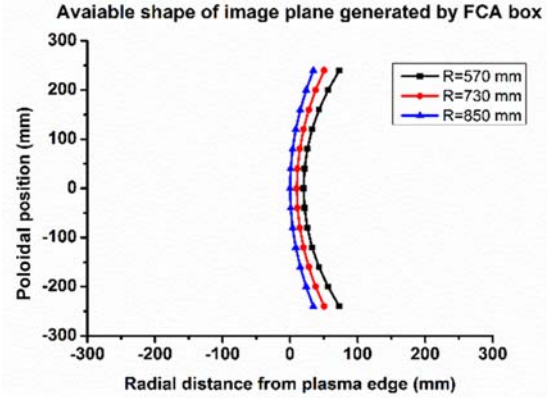


**Fig.5** Observation range of both two zoom options. Red line shows the wide option. Black line shows the narrow option

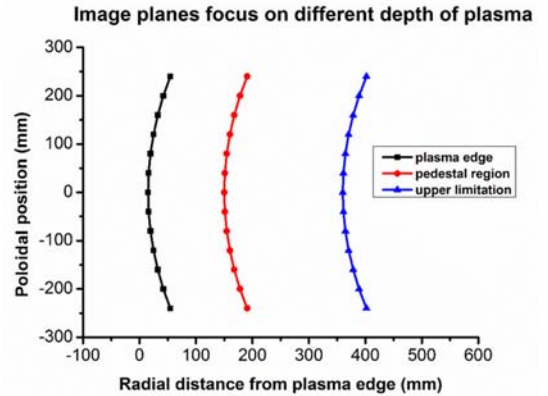
The FCA box is the key part of the receiver optics to generate a specified shape of image plane for perfect matching. The original idea of the FCA lenses is using meniscus type lenses to generate field curvature in the reversed direction and to balance the field curvature generated by other convex lenses. A couple of meniscus lenses have been used to reduce field curvature on the ECEI receiver optics on the J-TEXT device as the 1st generation FCA lenses [12]. According to the latest FCA technique, a new couple of lenses with optimized shape and surface radius are employed to adjust the shape of the image plane in a more flexible fashion, as shown in the green FCA box. The curvature of the image plane is modified by changing the distance between the couple of FCA lenses. These two FCA lenses reverse the curved direction of the image planes into negative radius. The available range of the image plane radius is from 570 mm to 850 mm generated by these couple of FCA lenses, which covers the whole radius range of the cutoff layers in the right-hand extraordinary mode operation mode, as shown in Fig. 6. It means that the FCA box is able to work out the perfectly matching image plane on the plasma side to improve image quality.

In the light blue Focus box, there is only one focus lens to adjust the image position in the radial direc-

tion. The image plane with a curved shape locked by a certain setup of the FCA box is able to move forward or backward in the plasma by moving the focus lens to minimize the off-focus distance. The available range of the image position is from normalized radius 0.18 to 0.99 as shown in Fig. 7, which is able to cover the plasma edge, pedestal region, and core region (upper limitation to normalized radius 0.2). The upper limitation for the MIR system is the position where the cutoff frequency equals the 2nd harmonic frequency of electron cyclotron emission.



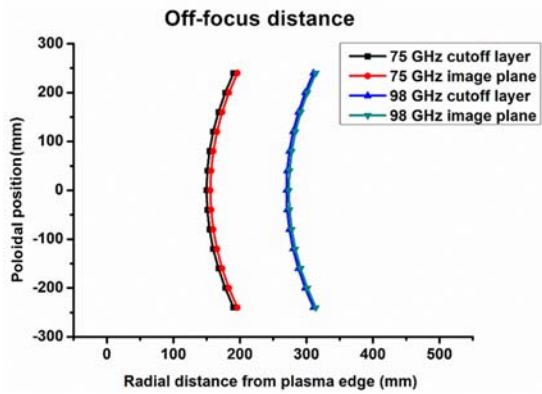
**Fig.6** Available shapes of image plane with different negative radii



**Fig.7** The image plane focus on different depths of plasma, including the plasma edge, pedestal region and upper limitation

According to the above simulation in the certain configuration, the receiver optics is able to generate a perfect image plane with negative radius to match the cutoff layer with less than 5 mm off-focus distance, which is much smaller than the 15 mm radial resolution. The cutoff layers and image planes for 75 GHz and 98 GHz illumination frequencies are shown in Fig. 8. The minimized off-focus distance gives a perfectly matching case for receiver optics imaging processing.

In the EAST 2015 campaign and the future, the toroidal magnetic field will be above 2.3 T, which is stronger than that in the DIII-D configuration. Thus, the image plane will be a little flatter than these simulation cases, which means the designed front-end optics system could fully meet the requirement of the MIR diagnostic on the EAST tokamak.



**Fig.8** The cutoff layers and image planes of characteristic frequencies, including 75 GHz cutoff layer (black), 75 GHz image plane (red), 98 GHz cutoff layer (blue) and 98 GHz image plane (green)

### 3 Summary

The goal of the EAST MIR system is to provide two dimensional density fluctuation imaging measurement capability with 96 channels (pixels). Both of the transmitter and receiver optics with large aperture are designed to match the cutoff layer. Three specific boxes have independent abilities to change poloidal spatial resolution, shape, and radial focus position, which provides a perfect matching image plane to increase image quality. The designed optics system is the first one for MIR diagnostic with field curvature adjustment lenses, which will be more flexible to match the target surface in different operation regions.

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